

TUPC Learning Commons Development

Installation of Requirements

System Requirements:

1. Python 3.9
 2. Flask
 3. MySQL Connector
 4. QR Code Library
 5. Visual Studio Code
 6. XAMPP
 7. Python VS Code Extension
 8. TUPC Learning Commons System Files
 9. VS Code Virtual Environment
 10. Setting up the Learning Commons Database
 11. Opening and Testing TUPC-LC System Files
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Part 1: Installing Python 3.9

1. Go to this website and download the .exe file
<https://www.python.org/ftp/python/3.9.0/python-3.9.0-amd64.exe>
2. Run the .exe file
3. Check/Tick the "Add Python to PATH" [IMPORTANT]
4. Click on "Install Now"

Note: Make sure that there are no other Python versions installed

Part 2: Installing Flask

1. Open Command Prompt/Terminal as administrator
2. Confirm the Python and pip install

```
python --version
python -m pip --version
```

3. Install Flask

```
python -m install flask
```

4. Verify Flask installation

```
python -m flask -version
```

Part 3: Installing MySQL Connector (Python)

1. Open Command Prompt/Terminal as administrator
2. Install MySQL Connector

```
python -m pip install mysql-connector-python
```

Part 4: Install QR Code Library (Python)

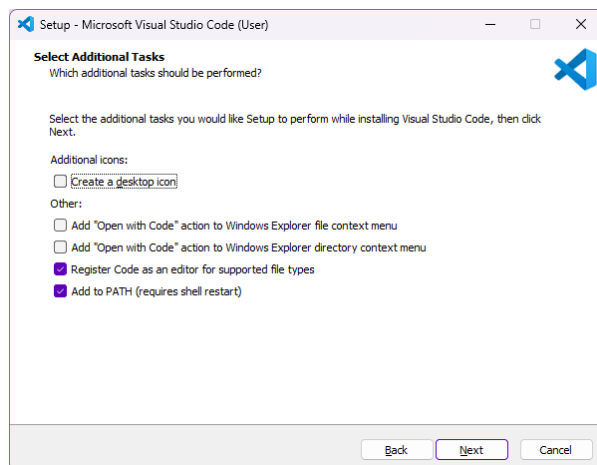
1. Open Command Prompt/Terminal as administrator
2. Install QR Code Library

```
pip install qrcode[pil]
```

Part 5: Installing Visual Studio Code

Note: You may skip this step if you already have VS Code installed

1. Open this website and download the .exe file
<https://code.visualstudio.com/sha/download?build=stable&os=win32-x64-user>
2. Click on "Next" until you've reached the "Select Additional Tasks" section of the install
3. Make sure that you have "Register Code as an editor for supported file types" and "Add to PATH (requires shell restart)".

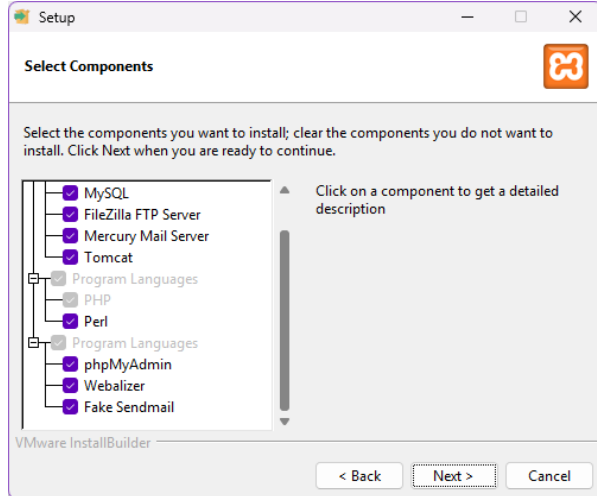


4. Click on "Install" and wait for the installation to finish
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Part 6: Installing XAMPP

1. Open this website and download the latest version
<https://www.apachefriends.org/download.html>
2. Run the installer

3. Make sure have the following components selected



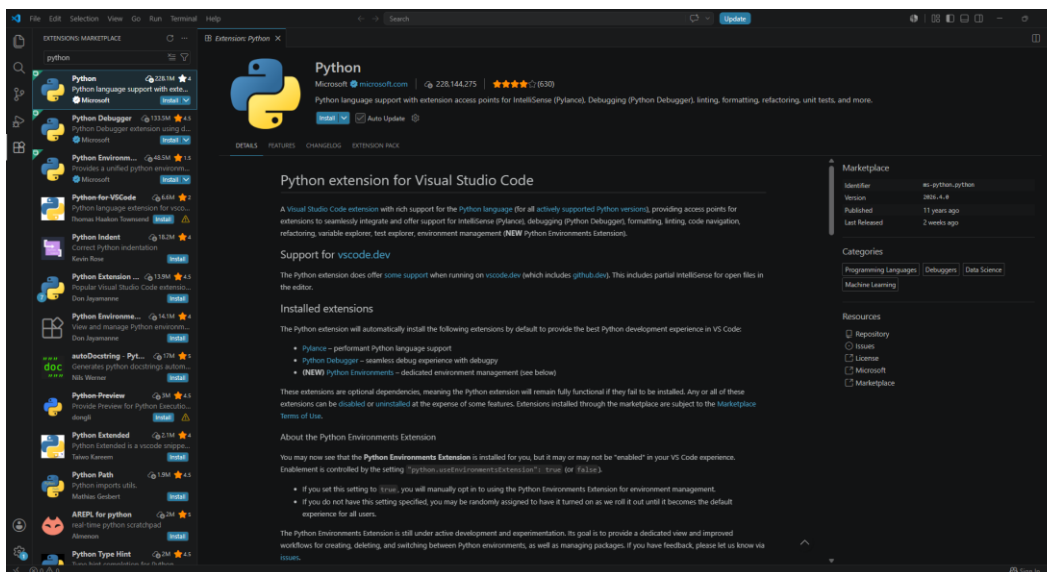
4. Keep the installation path as is (C:\xampp)

5. Click on "Next" and "Install"

6. Wait for the installation to finish

Part 7: Installing Python VS Code Extension

1. Open Visual Studio Code
2. Go to the "Extensions" tab
3. Search for "Python"
4. Select "Python" by Microsoft
5. Click on "Install"



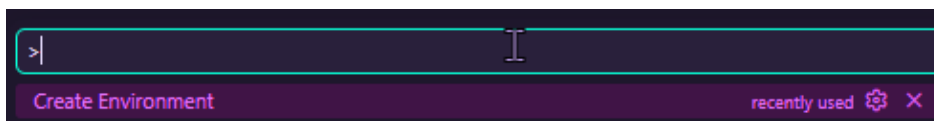
Part 8: Opening the TUPC Learning Commons System Files

1. Open this Google Drive link using your GSFE account to access the System Files folder
https://drive.google.com/drive/folders/17eISbiWlp6OGEYbU_985ZLCLyUFINKOy?usp=drive_link
2. Download the latest system update and extract the .zip file once downloaded
3. Open Visual Studio Code and open the TUPC Learning Commons folder that you've just downloaded

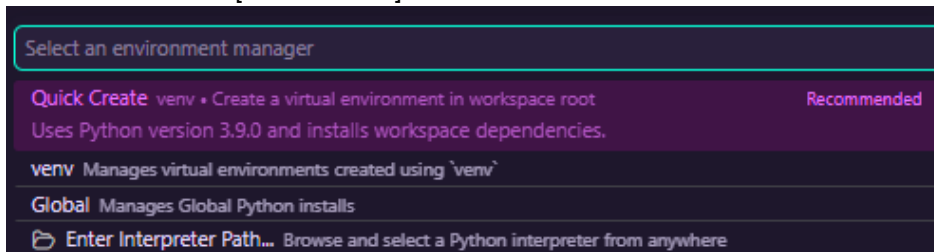
*Note: To open both admin and kiosk systems at the same time, make sure to go to File -> New Window -> *and select the folder you want to open*

Part 9: Creating a Virtual Environment

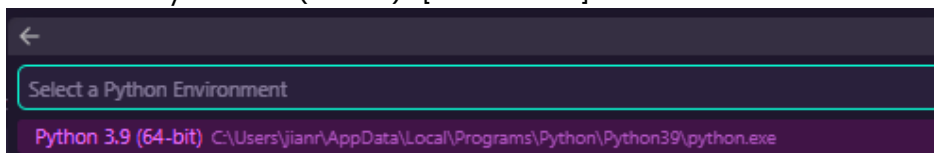
1. Open the TUPC Learning Commons folder in VS Code
2. Click the search bar located in the top part of the screen
3. Click on Ctrl + P
4. Click on "Create Environment"



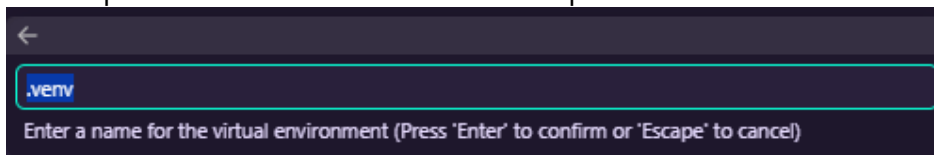
5. Select "venv" [IMPORTANT]



6. Select "Python 3.9 (64-bit)" [IMPORTANT]



7. Keep the environment name as is and press Enter



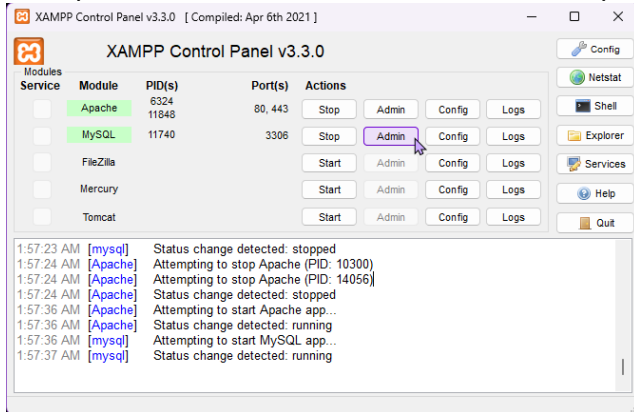
Important Note: Make sure to do this on all folders (tupclc_adminsystem, tupclc_kiosksystem, and tupclc_trackingssystem) of the TUPC Learning Commons

Part 10: Setting up the Learning Commons Database

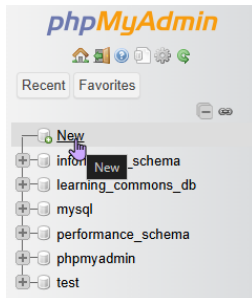
1. Open the TUPC Learning Commons System Files Google Drive Folder
2. Download “learning_commons_db.sql” file located on the main folder. You may also click the link below for the direct download.

<https://drive.google.com/file/d/10SsV2BEuWhVfRJG-pLaf70deW5M5GTZH/view?usp=sharing>

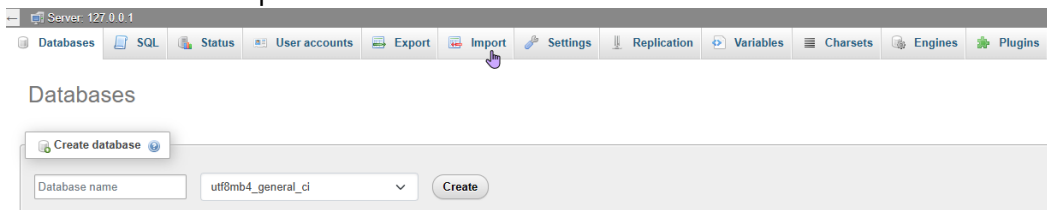
3. Open XAMPP and select “Admin” under MySQL.



4. On the left sidebar, click on “New”

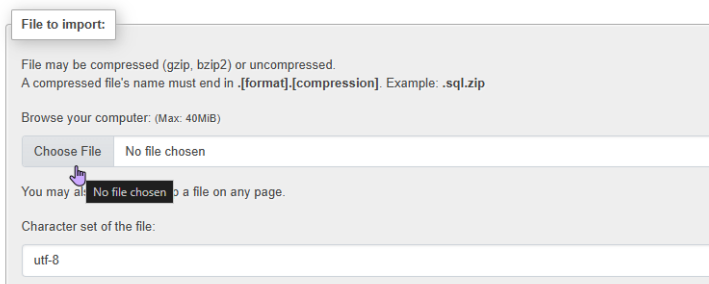


5. Look for the “Import” tab and click it.



6. Click on “Choose File” and look for “learning_commons_db.sql” file.

Importing into the current server



7. Scroll down and click on “Import”

Format-specific options:

SQL compatibility mode:

NONE

☒ Do not use AUTO_INCREMENT for zero values

Import

8. Check if the database has been imported. The database should look like this:

The screenshot shows the phpMyAdmin interface for a MySQL server. The left sidebar shows the database structure, with 'learning_commons_db' selected. The main panel displays the 'Structure' tab for the 'learning_commons_db' database. It shows a table list with columns: Table, Action, Rows, Type, Collation, Size, and Overhead. The tables listed are appointments, locations, restrictions, and stations. Below the table list, there is a 'Create new table' section with fields for 'Table name' and 'Number of columns' (set to 4), and a 'Create' button.

Table	Action	Rows	Type	Collation	Size	Overhead
appointments	Browse Structure Search Insert Empty Drop	8	InnoDB	utf8mb4_general_ci	48.0 K	-
locations	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 K	-
restrictions	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 K	-
stations	Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	32.0 K	-
4 tables	Sum	20	InnoDB	utf8mb4_general_ci	144.0 K	0 B

Part II: Opening and Testing TUPC-LC System Files

1. Open XAMPP Control Panel
2. Start “Apache” and “MySQL”
3. Open the TUPC Learning Commons folder
4. Look for app.py and click on the “Run Python File” button (look for the “play” button) located at the tabs section to open the VS Code’s built-in terminal
5. On the terminal below, type the following command

```
flask --app app.py --debug run
```

6. Press alt + click on <http://120.0.0.1:5000> to open the program and test it

```
PS D:\University\TUPC Learning Commons\tupclc_kiosksystem> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned)
(.venv) PS D:\University\TUPC Learning Commons\tupclc_kiosksystem> flask --app app.py --debug run
* Serving Flask app 'app.py'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 445-626-258
```

Admin System Login Password

Username: admin

Password: 1234

Notes:

- You may skip Steps 1 to 10 if you are opening the system files for the second time onwards since it is only a one-time procedure.
- To open a new system file (from kiosk to admin and vice versa), press Ctrl + C to close the debugger and run the flask command on the other folder. The debugger can only open once app.py file at a time.
- Some system features require the usage of a QR code scanner hardware.